

CHAPTER 1

$$1-1. (a) \quad C(s) = G_1(s) E(s) + G_2(s) E(s) = [G_1(s) + G_2(s)] E(s)$$

$$\therefore \frac{C(s)}{E(s)} = \underline{G_1(s) + G_2(s)}$$

$$(b) \quad C(s) = G_2(s) M(s) = G_1(s) G_2(s) \quad \therefore \frac{C(s)}{E(s)} = \underline{G_1(s) G_2(s)}$$

$$1-2. (a) \quad C(s) = G_p(s) M(s) = G_c(s) G_p(s) E(s) = G_c(s) G_p(s) [R(s) - H(s) C(s)]$$

$$[1 + G_c(s) G_p(s) H(s)] C(s) = G_c(s) G_p(s) R(s)$$

$$\therefore \frac{C(s)}{R(s)} = \underline{\frac{G_c(s) G_p(s)}{1 + G_c(s) G_p(s) H(s)}}$$

$$(b) \quad C(s) = G_3(s) M(s) = G_3(s) [G_1(s) E(s) + G_2(s) R(s)]$$

$$= G_3(s) G_1(s) [R(s) - H(s) C(s)] + G_2(s) G_3(s) R(s)$$

$$[1 + G_1(s) G_3(s) H(s)] C(s) = [G_1(s) + G_2(s)] G_3(s) R(s)$$

$$\therefore \frac{C(s)}{R(s)} = \underline{\frac{[G_1(s) + G_2(s)] G_3(s)}{1 + G_1(s) G_3(s) H(s)}}$$

$$(c) \quad C(s) = G_2(s) M(s) = G_1(s) G_2(s) E(s) = G_1(s) G_2(s) [R(s) - H_2(s) M(s) - H_1(s) C(s)]$$

$$\text{and } M(s) = C(s)/G_2(s)$$

$$\therefore \left[1 + \frac{G_1 G_2 H_2}{G_2} + G_1 G_2 H_1 \right] C(s) = G_1 G_2 R(s)$$

$$\therefore \frac{C(s)}{R(s)} = \underline{\frac{G_1(s) G_2(s)}{1 + G_1(s) H_2(s) + G_1(s) G_2(s) H_1(s)}}$$

$$1-3. (a) \quad \frac{C(s)}{R(s)} = \frac{G_c(s) G_p(s)}{1 + G_c(s) G_p(s) H(s)} \quad (b) \quad \frac{C(s)}{R(s)} = \frac{G_1(s) G_3(s) + G_2(s) G_3(s)}{1 + G_1(s) G_3(s) H(s)}$$

$$(c) \quad \frac{C(s)}{R(s)} = \frac{G_1(s) G_2(s)}{1 + G_1(s) H_2(s) + G_1(s) G_2(s) H_1(s)}$$

$$1-4. (a) \quad G_p(s) = \frac{C(s)}{M(s)} = \frac{5}{0.2s+1} = \frac{25}{s+5} \Rightarrow (s+5)C(s) = 25M(s)$$

$$\therefore \underline{\frac{dC(t)}{dt} + 5C(t) = 25m(t)}$$

$$1-4.(b) \quad m(t) = 10e(t) = 10[r(t) - c(t)]$$

$$\therefore \dot{c}(t) + 5c(t) = 250[r(t) - c(t)]$$

$$\therefore \underline{\dot{c}(t) + 255c(t) = 250r(t)}$$

$$(c) \quad 5C(s) + 255C(s) = 250R(s)$$

$$\therefore \frac{C(s)}{R(s)} = \frac{250}{s + 255}$$

$$(d) \quad \frac{C(s)}{R(s)} = \frac{(10) \frac{25}{s+5}}{1 + (10) \frac{25}{s+5} (1)} = \frac{250}{s + 255}$$

$$(e) \text{ open-loop: } \tau = 0.2s$$

$$\text{closed-loop: } \tau = 1/255 = 0.00392s$$

$$15.(a) \quad G(s) = \frac{C(s)}{M(s)} = \frac{3s+8}{s^2+2s+2}$$

$$\therefore (s^2+2s+2)C(s) = (3s+8)M(s)$$

$$\underline{\ddot{c}(t) + 2\dot{c}(t) + 2c(t) = 3\dot{m}(t) + 8m(t)}$$

$$(b) \quad m(t) = 2e(t) = 2[r(t) - c(t)]$$

$$\therefore \ddot{c}(t) + 2\dot{c}(t) + 2c(t) = 6[\dot{r}(t) - \dot{c}(t)] + 16[r(t) - c(t)]$$

$$\therefore \underline{\ddot{c}(t) + 8\dot{c}(t) + 18c(t) = 6\dot{r}(t) + 16r(t)}$$

$$(c) \quad (s^2+8s+18)C(s) = (6s+16)R(s)$$

$$\therefore \frac{C(s)}{R(s)} = \frac{6s+16}{s^2+8s+18}$$

$$(d) \quad \frac{C(s)}{R(s)} = \frac{(2) \frac{3s+8}{s^2+2s+2}}{1 + (2) \frac{3s+8}{s^2+2s+2} (1)} = \frac{6s+16}{s^2+8s+18}$$

$$(e) \text{ open-loop: } s^2+2s+2 \Rightarrow s = -1 \pm j$$

$$\therefore \text{term} = A_1 e^{-t} \cos(t + \theta_1); \tau = 1/1 = 1s$$

$$\text{closed-loop: } s^2+8s+18 \Rightarrow s = -4 \pm j\sqrt{2}$$

$$\therefore \text{term} = A_2 e^{-4t} \cos(\sqrt{2}t + \theta_2); \tau = 1/4 = 0.25s$$

$$1-6.(a) \quad G(s) = \frac{C(s)}{M(s)} = \frac{5}{s^2+2s+2} \Rightarrow (s^2+2s+2)C(s) = 5M(s)$$

$$\therefore \underline{\ddot{c}(t) + 2\dot{c}(t) + 2c(t) = 5m(t)}$$

$$1-6.(b) \quad E(s) = R(s) - (3s+1)C(s)$$

$$\therefore e(t) = r(t) - 3\dot{c}(t) - c(t)$$

$$\therefore \ddot{c}(t) + 2\dot{c}(t) + 2c(t) = 5[2e(t)] = 10r(t) - 30\dot{c}(t) - 10c(t)$$

$$\therefore \underline{\ddot{c}(t) + 32\dot{c}(t) + 12c(t) = 10r(t)}$$

$$(c) \quad (s^2 + 32s + 12)C(s) = 10R(s)$$

$$\therefore \underline{\frac{C(s)}{R(s)} = \frac{10}{s^2 + 32s + 12}}$$

$$(d) \quad \frac{C(s)}{R(s)} = \frac{(2) \frac{5}{s^2 + 2s + 2}}{1 + (2) \frac{5}{s^2 + 2s + 2} (3s+1)} = \underline{\frac{10}{s^2 + 32s + 12}}$$

$$(e) \quad \text{open-loop: } \gamma = 1s, \text{ from Problem 1-5(e)}$$

$$\text{closed-loop: poles} = -16 \pm \sqrt{244} = -31.62, -0.38$$

$$\therefore \tau_1 = 1/31.62 = \underline{0.0316s}; \tau_2 = 1/0.38 = \underline{2.63s}$$

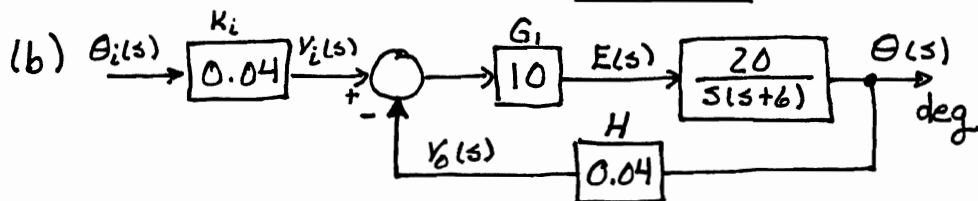
$$1-7.(a) \quad \Theta(s) = \frac{20}{s^2(s+6)} = \frac{3.33}{s^2} + \frac{b}{s} + \frac{5/9}{s+6}$$

$$b = \frac{d}{ds} \left[\frac{20}{s+6} \right]_{s=0} = \frac{-20}{(s+6)^2} \Big|_{s=0} = -\frac{5}{9}$$

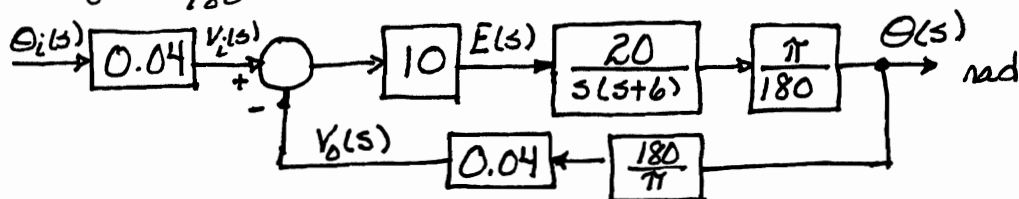
$$\therefore \underline{\Theta_{ss}(t) = (3.33t - \frac{5}{9}) \text{ in degrees}}$$

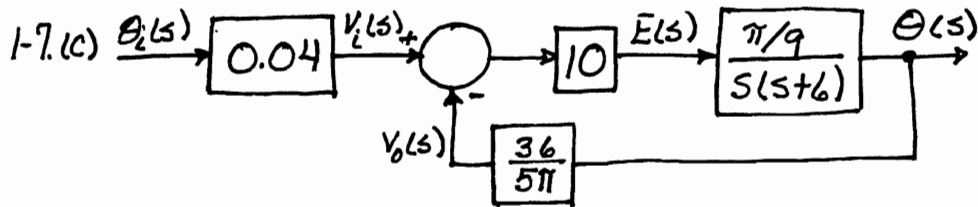
$$\dot{\Theta}_{ss}(t) = \underline{3.33 \text{ deg/s}}$$

$$\therefore 3.33 \frac{d}{s} \times \frac{60s}{1 \text{ min}} \times \frac{1 \text{ rev}}{360^\circ} = \underline{\frac{5}{9} \text{ rpm}}$$



$$(c) \quad 1 \text{ degree} = \frac{\pi}{180} \text{ rad}$$





$$(d) \frac{\Theta(s)}{\Theta_i(s)} = \frac{(0.04)(10) \frac{20}{s(s+6)}}{1 + (0.04)(10) \frac{20}{s(s+6)}} = \frac{8}{s^2 + 6s + 8}$$

$$\Theta(s) = \frac{8}{(s+2)(s+4)} \times \frac{10}{s} = \frac{10}{s} + \frac{-20}{s+2} + \frac{10}{s+4}$$

$$\therefore \Theta(t) = 10 - 20e^{-2t} + 10e^{-4t}, \quad t \geq 0$$

$$(e) e^{-t/\tau} \Rightarrow \frac{1}{\tau_1} = 2, \tau_1 = 0.5, \tau_2 = 1/4 = 0.25, \therefore t_{ss} \approx 4(0.5) = 2s$$

$$1-8. (1-15) \frac{K_T/JR_a}{s(s + \frac{BR_a + K_T K_b}{JR_a})} = \frac{20}{s(s+6)}, \therefore \frac{K_T}{JR_a} = 20; \frac{BR_a + K_T K_b}{JR_a} = 6$$

$$(1-16), (1-17), \quad x_1 = \Theta; \quad x_2 = \dot{\Theta} = \dot{x}_1$$

$$(1-17) \quad \dot{x}_2(t) = \ddot{\Theta}(t) = -6x_2(t) + 20e(t) \\ = -6x_2(t) + 20(10[0.04\Theta_i(t) - 0.04\Theta(t)]) \\ = -6x_2(t) + 8\Theta_i(t) - 8x_1(t)$$

$$\therefore \dot{\underline{x}}(t) = \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix} \underline{x}(t) + \begin{bmatrix} 0 \\ 8 \end{bmatrix} \Theta_i(t)$$

$$\Theta(t) = [1 \quad 0] \underline{x}(t)$$

$$1-9. (a) \text{ From Problem 1-7(d), } \frac{\Theta(s)}{\Theta_i(s)} = \frac{8}{s^2 + 6s + 8}$$

$$(b) 1 \text{ deg} = (\pi/180) \text{ rad}$$

$$\therefore \frac{\Theta(s)}{\Theta_i(s)} = \frac{8}{s^2 + 6s + 8} \times \frac{\pi}{180} = \frac{2\pi/45}{s^2 + 6s + 8}$$

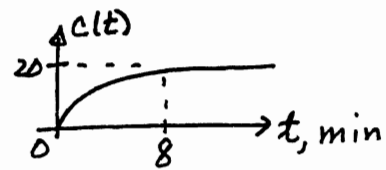
$$(c) \frac{\Theta(s)}{\Theta_i(s)} = \frac{K_i G_1 G_2}{1 + G_1 G_2 H} = \frac{(\frac{1}{25})(10)(\frac{\pi/9}{s(s+6)})}{1 + (10)(\frac{\pi/9}{s(s+6)})(\frac{36}{5\pi})} = \frac{2\pi/45}{s^2 + 6s + 8}$$

$$1-10.(a) \quad G(s) = \frac{2}{s+0.5} = \frac{K}{\tau s + 1} = \frac{K/\tau}{s+1/\tau} ; \therefore \tau = \underline{2 \text{ min}}$$

$$(b) \quad C(s) = \frac{2}{s+0.5} \times \frac{5}{s} = \frac{20}{s} + \frac{-20}{s+0.5}$$

$$\therefore c(t) = \underline{20(1 - e^{-0.5t})}, t \geq 0$$

$$c_{ss}(t) = \underline{20^\circ\text{C}}$$



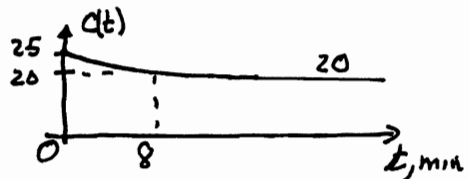
$$(c) \quad \frac{C(s)}{E(s)} = \frac{2}{s+0.5}$$

$$\therefore \dot{c}(t) + 0.5c(t) = 2e(t)$$

$$sC(s) - c(0) + 0.5C(s) = 2E(s)$$

$$\therefore C(s) = \frac{2E(s)}{s+0.5} + \frac{c(0)}{s+0.5} \Rightarrow c(t) = \overbrace{20(1 - e^{-0.5t})}^{\text{from (b)}} + 25e^{-0.5t}$$

$$\therefore c(t) = \underline{20 + 5e^{-0.5t}}, t \geq 0$$



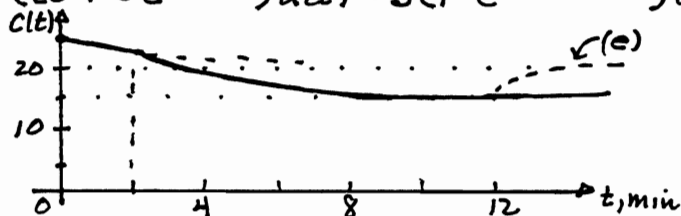
(d) Disturbance response:

$$C_d(s) = \frac{-2.5}{s+0.5} \cdot \frac{1}{s} = \frac{-5}{s} + \frac{5}{s+0.5}$$

$$\therefore c_d(t) = \underline{-5(1 - e^{-0.5(t-2)})}u(t-2), \text{ since door opened at } t_0=2.$$

$\therefore$ from (c) and (d),

$$c(t) = (20 + 5e^{-0.5t})u(t) - 5(1 - e^{-0.5(t-2)})u(t-2)$$



$$(e) \quad c(12) = 20 + 5e^{-6} - 5 + 5e^{-5} \approx \underline{15}$$

From (c), with $c(0) = 15$

$$c_1(t) = 20(1 - e^{-0.5t}) + 15e^{-0.5t} = 20 - 5e^{-0.5t}$$

$$\therefore \underline{c(t) = [20 - 5e^{-0.5(t-12)}]u(t-12), t \geq 12}$$

$$1-11. (a) \quad \tau = 2 \text{ min} \rightarrow 120 \text{ s}$$

$$\therefore \frac{C(s)}{E(s)} = \frac{K}{\tau s + 1} = \frac{4}{2s + 1} \Rightarrow \frac{4}{120s + 1} = \frac{0.0333}{s + 0.00833}$$

$$(b) \text{ (i) From Problem 1-4(b), } c(t) = 20(1 - e^{-0.5t}), t \geq 0$$

$$\therefore C(1) = 20(1 - e^{-0.5})$$

$$\text{(ii) } C(s) = \frac{0.0333}{s + 0.00833} \times \frac{5}{s} = \frac{20}{s} + \frac{-20}{s + 0.00833}$$

$$\therefore c(t) = 20(1 - e^{-0.00833t}), t \geq 0$$

$$C(1) = 20(1 - e^{-0.00833(60)}) = 20(1 - e^{-0.5})$$

$$1-12. (a) \quad \Theta(s) = \frac{1}{Js^2} T(s) = \frac{K}{Js^2} E(s) = \frac{K}{Js^2} [\Theta_c(s) - \Theta(s)]$$

$$\therefore [1 + \frac{K}{Js^2}] \Theta(s) = \frac{K}{Js^2} \Theta_c(s)$$

$$\therefore \frac{\Theta(s)}{\Theta_c(s)} = \frac{K/Js^2}{1 + \frac{K}{Js^2}} = \frac{K/J}{s^2 + K/J}$$

$$(b) \quad (1-4): x_1 = 0; \quad (1-5): x_2 = \dot{\theta} = \dot{x}_1$$

$$\therefore \dot{x}_2(t) = \ddot{\theta}(t) = \frac{1}{J} \tau(t) = \frac{K}{J} e(t) = \frac{K}{J} [\theta_c(t) - \theta(t)]$$

$$= \frac{K}{J} [\theta_c(t) - x_1(t)]$$

$$\therefore \begin{bmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ K/J & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} + \begin{bmatrix} 0 \\ K/J \end{bmatrix} \theta_c(t)$$

$$y(t) = [1 \quad 0] \underline{x}(t)$$

$$1-13. (a) \text{ From Problem 1-12, } \frac{\Theta(s)}{\Theta_c(s)} = \frac{36}{s^2 + 36}$$

$$\therefore \Theta(s) = \frac{36}{s^2 + 36} \times \frac{20}{s} = \frac{720}{s(s^2 + 36)} = \frac{20}{s} + \frac{as + b}{s^2 + 36}$$

$$= \frac{20s^2 + 720 + as^2 + bs}{s(s^2 + 36)}; \therefore b = 0, a = -20$$

$$\therefore \Theta(s) = \frac{20}{s} + \frac{-20s}{s^2 + 36} \Rightarrow \theta(t) = 20[1 - \cos 6t], t \geq 0$$

1-13. (b) From (a), $\ddot{\theta}(t) + 36\theta(t) = 36\theta_c(t)$

$$\therefore (s^2 + 36)\theta(s) - s\theta(0) - \dot{\theta}(0) = 36\theta_c(s)$$

$$\therefore \theta(s) = \frac{36}{s^2 + 36}\theta_c(s) + \frac{10s}{s^2 + 36} + \frac{30}{s^2 + 36}$$

$$= \frac{20}{s} - \frac{20s}{s^2 + 36} + \frac{10s}{s^2 + 36} + \frac{30}{s^2 + 36}$$

$$= \frac{20}{s} - \frac{10s}{s^2 + 36} + \frac{30}{s^2 + 36}$$

$$\therefore \theta(t) = \underline{20 - 10 \cos 6t + 5 \sin 6t}, \quad t \geq 0$$

(c) $\theta(0) = 20 - 10 = 10^\circ$

$$\dot{\theta}(t) = 60 \sin 6t + 30 \cos 6t \Rightarrow \dot{\theta}(0) = \underline{30^\circ/s}$$

From (b), $\ddot{\theta} + 36\theta = 720$

$$\therefore 360 \cos 6t - 180 \sin 6t + 720 - 360 \cos 6t + 180 \sin 6t = 720$$

$$\therefore \underline{720 = 720}$$

1-14. From Problem 1-12,

$$\theta(s) = \frac{K/J}{s^2 + K/J} \times \frac{5}{s} = \frac{a^2 5}{(s^2 + a^2)s}, \quad a^2 = K/J$$

$$\therefore \theta(s) = \frac{5}{s} - \frac{5s}{s^2 + a^2}$$

$$\therefore \theta(t) = \underline{5 - 5 \cos at}, \quad t \geq 0$$

$$10 \text{ cy/min} \times \frac{1 \text{ min}}{60 \text{ s}} = \frac{1}{6} \text{ cy/s} \Rightarrow 2\pi\left(\frac{1}{6}\right) \text{ rad/s} = a$$

$$\therefore a = \pi/3 = \sqrt{K/J}, \therefore K \text{ and } J \text{ cannot be determined without additional data.}$$

1-15 (a) $\theta(s) = \frac{1}{s} \dot{\theta}(s) = \frac{1}{Js^2} T(s) = \frac{K}{Ts^2} [\theta_c(s) - K_r \dot{\theta}(s) - \theta(s)]$

$$\dot{\theta}(s) = s\theta(s)$$

$$\therefore \theta(s) = \frac{K}{Js^2} [\theta_c(s) - (K_r s + 1)\theta(s)]$$

$$1-15.(a) \therefore \frac{\Theta(s)}{\Theta_c(s)} = \frac{K}{Js^2 + Kk_r s + K} = \frac{K/J}{s^2 + \frac{Kk_r}{J}s + \frac{K}{J}}$$

$$(b) \quad x_1(t) = \theta(t); \quad x_2(t) = \dot{x}_1(t) = \dot{\theta}(t)$$

$$\therefore \dot{x}_2(t) = \ddot{\theta}(t) = \frac{1}{J} \tau(t) = \frac{K}{J} [\theta_c(t) - k_r x_2(t) - x_1(t)]$$

$$\therefore \dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -k/J & -Kk_r/J \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ K/J \end{bmatrix} \theta_c(t)$$

$$\theta(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t)$$

$$(c) \quad |sI - A| = \begin{vmatrix} s & -1 \\ K/J & s + Kk_r/J \end{vmatrix} = s^2 + \frac{Kk_r}{J}s + \frac{K}{J}$$

$$1-16. \quad \theta_a(s) = \frac{1}{100} \theta_m(s) \quad \frac{1}{100s} \dot{\theta}_m(s) = \frac{2}{s(0.5s+1)} E_a(s) = \frac{2K}{s(0.5s+1)} M(s)$$

$$\therefore \frac{\theta_a(s)}{M(s)} = \frac{2K}{s(0.5s+1)}; \quad \frac{\theta_a(s)}{E_a(s)} = \frac{2}{s(0.5s+1)}$$

$$1-17.(a) \quad \dot{\theta}_a(s) = \frac{2}{0.5s+1} \times \frac{24}{s} = \frac{48}{s(0.5s+2)} = \frac{96}{s(s+2)} = \frac{k_1}{s} + \frac{k_2}{s+2}$$

$$= \frac{48}{s} - \frac{48}{s+2} \Rightarrow \dot{\theta}_a(t) = 48 - 48e^{-2t}, \quad t \geq 0$$

$$\therefore \dot{\theta}_{ass}(t) = 48^\circ/s \times \frac{60s}{1min} \times \frac{1rev}{360^\circ} = 8 \text{ rpm}$$

$$\dot{\theta}_{mss}(t) = 100 \dot{\theta}_a(t) = 800 \text{ rpm}$$

$$(b) \text{ From (a), } \dot{\theta}_a(t) = 48^\circ/s$$

$$(c) \text{ From (a), } \dot{\theta}_a(t) = 48(1 - e^{-2t}), \quad t \geq 0$$

$$\therefore \tau = 0.5s$$

$$e^{-2t_1} = 0.01 \Rightarrow 2t_1 = 4.60 \Rightarrow t_1 = 2.30s$$

$$(d) \quad K = 2.4 \Rightarrow E_a(t) = 24V, \text{ rated voltage}$$